

Research Notes

Paper Airplane Aerodynamics Using a Modular Wind Tunnel

Fundamentals of Aerodynamics Chapter 2 Notes

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Project:

Experimental Investigation of Paper Airplane Aerodynamics Using a Modular Wind Tunnel

Why Aerodynamics Matters

Aerodynamics is the study of how air moves around an object.

For this project the object is a **paper airplane**.

Understanding airflow allows engineers to design airplanes that

- fly farther
- fly straighter
- create less drag
- produce more lift
- become more energy efficient

Modern aircraft, spacecraft and even Formula 1 cars all rely on these same aerodynamic principles.

Three Fundamental Laws of Aerodynamics

Everything in this project is based on three conservation laws.

1. Conservation of Mass

Mass cannot be created or destroyed.

As air enters the wind tunnel, the amount of air leaving must equal the amount entering.

This is represented mathematically by the **Continuity Equation**.

2. Conservation of Momentum

Newton's Second Law applies to airflow.

When air changes direction or slows down after passing over the airplane, the airplane experiences aerodynamic forces.

These forces become

- Lift
- Drag

Momentum equations are used to calculate these forces.

3. Conservation of Energy

Energy is not lost.

Instead, it changes form.

Examples include

- Pressure Energy
- Kinetic Energy
- Internal Energy

These relationships are described by the Energy Equation.

Lift

Lift is the upward aerodynamic force.

Lift is generated because

- air accelerates over the top surface
- pressure decreases
- pressure underneath is relatively higher

This pressure difference produces an upward force.

Paper airplanes with better wing geometry generally generate more lift.

Drag

Drag is the force resisting motion.

Every airplane creates a wake behind it.

Inside the wake,

- airflow slows
- momentum decreases
- turbulence increases

The drag force can be determined by measuring the velocity loss in this wake using a wind tunnel.

Wake

A wake is the disturbed airflow behind the airplane.

Important observations

Large wake

↓

Large energy loss

↓

Higher drag

Small wake

↓

Lower drag

↓

Better aerodynamic efficiency

Velocity Profiles

Air entering the wind tunnel has nearly uniform velocity.

After passing the airplane, velocity changes because the airplane disturbs the flow.

These velocity profiles are measured downstream to calculate drag.

Streamlines

A streamline shows the direction the air is moving at one instant.

If streamlines remain smooth

Flow is attached.

If streamlines separate

Turbulence forms.

Drag increases.

Pathlines

A pathline tracks the path of one air particle through time.

Think of following one tiny dust particle through the wind tunnel.

Streaklines

A streakline shows every air particle that has passed through one point.

If smoke is injected into the tunnel continuously,

the smoke creates a streakline.

For steady airflow,

- Streamlines
- Pathlines
- Streaklines

are identical.

Boundary Layer

As air touches the paper airplane,
friction slows the airflow.

This thin region is called the **boundary layer**.

Boundary layer behavior strongly affects

- drag
- flow separation
- stall

Control Volume

Instead of studying every air molecule,
engineers imagine an invisible box around the airplane.

This box is called a **control volume**.

Air enters,
moves around the airplane,
then exits.

Using this control volume,
engineers calculate

- Lift
 - Drag
 - Momentum change
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Why Wind Tunnels Are Used

Wind tunnels allow

- constant airspeed
- repeatable experiments
- accurate measurements
- visualization of airflow

Instead of flying the airplane repeatedly,
the airplane stays still while air moves around it.

Measurements Collected in This Research

The modular wind tunnel will measure

- Air velocity
 - Temperature
 - Wake velocity
 - Airflow uniformity
 - Drag trends
 - Lift observations
 - Smoke visualization
 - Flow separation
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Experimental Variables

Independent Variables

- Paper airplane design
 - Wing span
 - Wing sweep
 - Nose weight
 - Angle of attack
 - Airspeed
-

Dependent Variables

- Flight stability
 - Lift
 - Drag
 - Wake size
 - Glide distance
-

Instruments

The wind tunnel will use

- Airflow meter (anemometer)
 - Digital calipers
 - Smoke generator
 - Camera
 - Ruler
 - Force scale (future addition)
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Why This Research Matters

This project demonstrates that inexpensive laboratory equipment can be used to investigate the same aerodynamic principles employed in aircraft and spacecraft design. By constructing a modular wind tunnel and measuring airflow around paper airplanes, the study connects theoretical concepts, such as conservation of mass, momentum, energy, lift, drag, and wake formation, to experimental observations. The project provides hands-on experience with experimental methods commonly used in aerospace engineering and forms a foundation for future research involving computational fluid dynamics (CFD), advanced airfoil testing, and propulsion systems.